UNIVERSITY OF TEXAS ARLINGTON
COLLEGE OF SCIENCE

Advanced Classical Mechanics

Lecture 8: The Calculus of Variations

Readings: Morin - Chapter 6, Marion Chapter 6

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The statement of the problem

A historical perspective: The development of the calculus of variations was begun by Newton (1686) and was extended by Johann and Jakob Bernoulli (1696) and by Euler (1744). Adrien Legendre (1786), Joseph Lagrange (1788), Hamilton (1833), and Jacobi (1837) all made important contributions. The names of Peter Dirichlet (1805-1859) and Karl Weierstrass (1815-1879) are particularly associated with the establishment of a rigorous mathematical foundation for the subject.

Why do we need Calculus of Variations? Many problems in Newtonian mechanics are more easily analyzed by means of alternative statements of the laws, including **Lagrange's equation** and **Hamilton's principle**. As a prelude to these techniques, we consider in this lecture some general principles of the techniques of the calculus of variations.

What do we aim to find? Our primary interest here is in determining the path that gives extremum solutions, for example, the shortest distance (or time) between two points. A well-known example of the use of the theory of variations is **Fermat's principle**: Light travels by the path that takes the least amount of time.

The problem statement: The basic problem of the calculus of variations is to determine the function $y(x)$ such that the integral,

$$J = \int_{x_1}^{x_2} f(y(x), y'(x); x)\,dx\,, \qquad Eq\ 1$$

is an extremum (i.e., either a maximum or a minimum).

The statement of the problem

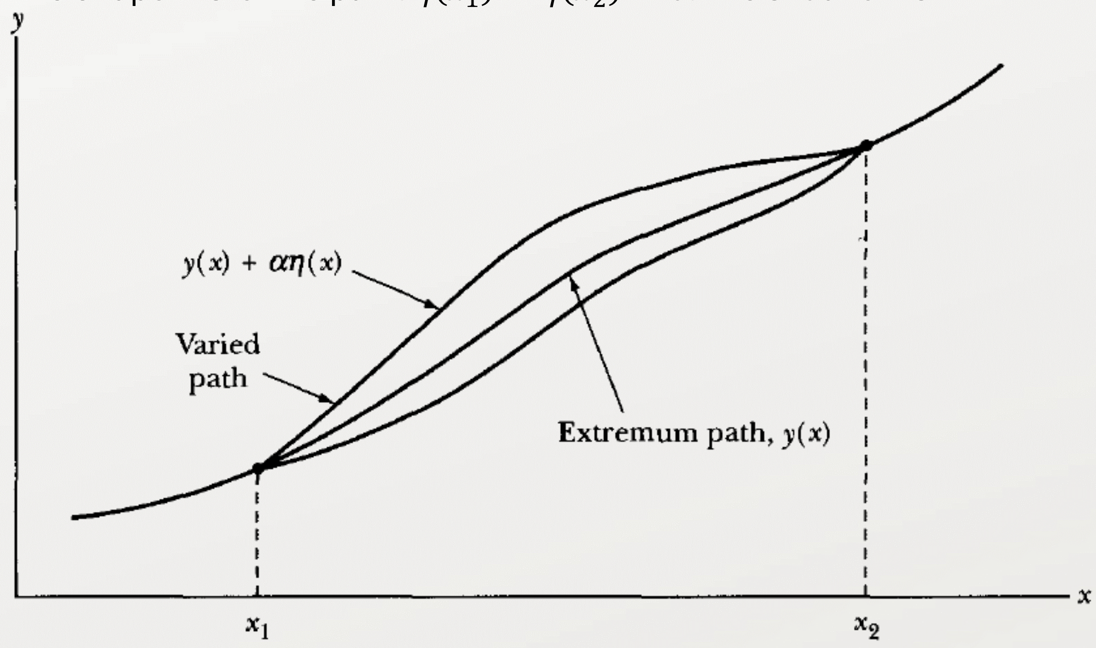
In the equation above, $y'(x) = dy/dx$, and the semicolon in $f(\cdot)$ separates the independent variable x from the dependent variable $y(x)$ and its derivative $y'(x)$. The integral quantity J is a generalization of a function called a **fanctional**, actually the **integral of a functional** in this case. The limits of integration do not have to be considered fixed. If they are allowed to vary, the problem increases to finding not only $y(x)$ but also x_1 and x_2 such that J is an extremum.

The functional J depends on the function $y(x)$, and the limits of integration are fixed. The function $y(x)$ is then to be varied until an extreme value of J is found. By extremization, we mean that if a function $y = y(x)$ makes the integral J a minimum value, then any neighboring function, no matter how close to $y(x)$, must increase the value of J.

The definition of a **neighboring function** may be made as follows: We give all possible functions $y(x)$ a parametric representation $y = y(\alpha, x)$ such that, for $\alpha = 0$, $y = y(0, x) = y(x)$ is the function that yields an extremum for J.
We can then write,

$$y(\alpha, x) = y(0, x) + \alpha\eta(x) \quad , \qquad Eq\ 2$$

where $\eta(x)$ is some function of x that has a continuous first derivative and that vanishes at x_1 and x_2, because the varied function must $y(\alpha, x)$ must be identical with $y(x)$ at the endpoints of the path: $\eta(x_1) = \eta(x_2) = 0$. The situation is depicted schematically in the figure.



The function $y(x)$ is the path that makes the functional J an extremum. The neighboring functions $y(x) + \alpha\eta(x)$ vanish at the endpoints and may be close to $y(x)$, but are not the extremum.

Euler's Equation

To determine the result of the condition expressed by Equation 4, we must differentiate Equation 3:

$$J(\alpha) = \int_{x_1}^{x_2} f(y(x), y'(x); x) dx\,, \qquad Eq\ 3$$

The condition that the integral have a **stationary value** (i.e., that an extremum results) is that J be independent of α in first order along the path giving the extremum ($\alpha = 0$), or, equivalently, that,

$$\frac{\partial J}{\partial \alpha} = 0\ \ , \qquad Eq\ 4$$

for all functions $\eta(x)$. Beware that this is only a **necessary condition**; but **not sufficient**.

Problem 1. (Linear Function Variation) Consider the function $f = \left(\frac{dy}{dx}\right)^2$, where $y(x) = x$. Add to $y(x)$ the function $\eta(x) = \sin(x)$, and find $J(\alpha)$ between the limits of $x = 0$ and $x = 2\pi$. Show that the stationary value of $J(\alpha)$ occurs for $\alpha = 0$.

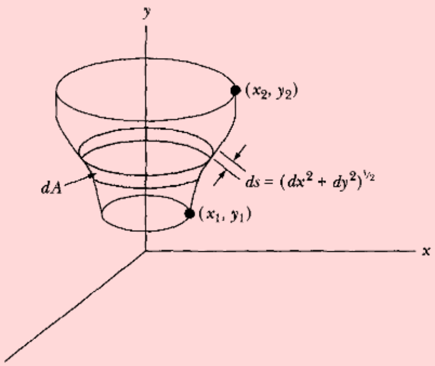
Problem 2. (Euler's Equation) To determine the result of the condition expressed by Equation 4, we must differentiate Equation 3. Show that this differentiation leads to the following differential equation that is known as the **Euler's Equation**, which expresses the necessary condition for the functional J to have an extremum. When applied to mechanical systems, this equation (first derived by Euler in 1744) is known as the **Euler-Lagrange equation**.

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0\ \ , \qquad Eq\ 5$$

Euler's Equation

Problem 3. (Brachistochrone) We can use the calculus of variations to solve a classic problem in the history of physics: the brachistochrone. Consider a particle moving in a constant force field starting at rest from some point (x_1, y_1) to some lower point (x_2, y_2). Find the path that allows the particle to accomplish the transit in the least possible time.

Problem 4. (Surface area minimization)
Consider the surface generated by revolving a line connecting two fixed points (x_1, y_1) and (x_2, y_2) about an axis coplanar with the two points. Find the equation of the line connecting the points such that the surface area generated by the revolution (i.e., the area of the surface of revolution) is a minimum.



Problem 5. (The second form of Euler's Equation) Sometimes, our functional $f(\cdot)$ does not **explicitly** depend on the independent variable x, that is, $\frac{\partial f}{\partial x} = 0$ for all x. Show that the Euler's Equation can be also written in a secondary form,

$$\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y' \frac{\partial f}{\partial y'}\right) = 0, \qquad Eq\ 6$$

which under the assumption of $\frac{\partial f}{\partial x} = 0$ becomes,

$$f - y' \frac{\partial f}{\partial y'} = \text{const.}, \qquad Eq\ 7$$

Problem 6. (Geodesic) A **geodesic** is a line that represents the shortest path between any two points when the path is restricted to a particular surface. Find the equation that describes a geodesic on a sphere.

Euler's Equation for functionals of several independent functions

The Euler equation derived in the preceding slides is the solution of the variational problem in which it was desired to find the single function $y(x)$ such that the integral of the functional f was an extremum. The case more commonly encountered in mechanics is that in which f is a functional of several dependent variables:

$$f = f(y_1(x), y_1'(x), y_2(x), y_2'(x), \dots; x) \equiv f(\{y_i(x), y_i'(x): i = 1,2, \dots, n\}; x), \qquad Eq\ 8$$

In analogy with Equation 2 and Problem 2. (Euler's Equation), we can again write,

$$y_i(\alpha, x) = y_i(0, x) + \alpha\eta_i(x) \quad , \qquad Eq\ 9$$

$$\frac{\partial J(\alpha)}{\partial \alpha} = \int_{x_1}^{x_2} \sum_{i=1}^{n} \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx}\frac{\partial f}{\partial y_i'} \right) \eta_i(x)\, dx\, , \qquad Eq\ 10$$

Because the individual variations (the $\eta_i(x)$) are all independent, the vanishing of the Equation 10 when evaluated at $\alpha = 0$ requires the separate vanishing of each expression in the brackets:

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx}\frac{\partial f}{\partial y_i'} = 0\ for\ i = 1, 2, \dots, n \ , \qquad Eq\ 10$$

which is a **generalized Euler Equation**.

Euler's Equation with auxiliary conditions

Suppose we want to find, for example, the shortest path between two points on a surface. Then, in addition to the conditions already discussed, there is the condition that the path must satisfy the equation of the surface, say, $g(y_i;\ x) = 0$. Such an equation was implicit in the solution of Problem 6 for the geodesic on a sphere where the condition was,

$$g = \sum x_i^2 - \rho = 0, \qquad Eq\ 8$$

which implies $r = \rho = \text{const.} \equiv \text{sphere radius.}$
In analogy with Equation 2 and Problem 2. (Euler's Equation), we can again write,

$$y_i(\alpha, x) = y_i(0, x) + \alpha\eta_i(x) \quad , \qquad Eq\ 9$$

$$\frac{\partial J(\alpha)}{\partial \alpha} = \int_{x_1}^{x_2} \sum_{i=1}^{n} \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx}\frac{\partial f}{\partial y_i'} \right) \eta_i(x)\, dx \,, \qquad Eq\ 10$$

But in the general case, we must make explicit use of the auxiliary equation or equations. These equations are also called **equations of constraint**. Consider the case in which

$$f = f(y, y', z, z'; x) \ , \qquad Eq\ 11$$

Then from Eq 10 we have,

$$\frac{\partial J(\alpha)}{\partial \alpha} = \int_{x_1}^{x_2} \left[\left(\frac{\partial f}{\partial y_i} - \frac{d}{dx}\frac{\partial f}{\partial y_i'} \right) \frac{\partial y}{\partial \alpha} + \left(\frac{\partial f}{\partial z_i} - \frac{d}{dx}\frac{\partial f}{\partial z_i'} \right) \frac{\partial z}{\partial \alpha} \right] dx \quad , \qquad Eq\ 12$$

But now there also exists an equation of constraint of the form, $g = g(y, z; x) = 0.$

Euler's Equation with auxiliary conditions

Thus, the variations $\frac{\partial y}{\partial \alpha}$ and $\frac{\partial z}{\partial \alpha}$ are no longer independent, so the expressions in parentheses in Equation 12 do not separately vanish at $\alpha = 0$. Differentiating g from Equation 11, we have,

$$dg = \left(\frac{\partial g}{\partial y}\frac{\partial y}{\partial \alpha} + \frac{\partial g}{\partial z}\frac{\partial z}{\partial \alpha}\right) d\alpha = 0$$

where no term in x appears since $\frac{\partial x}{\partial \alpha} = 0$. Now,

$$\left.\begin{aligned} y(\alpha, x) &= y(x) + \alpha\eta_1(x) \\ z(\alpha, x) &= z(x) + \alpha\eta_2(x) \end{aligned}\right\}$$

Therefore, by determining $\frac{\partial y}{\partial \alpha}$ and $\frac{\partial z}{\partial \alpha}$ from the above equation and inserting into the term in parentheses of the top equation, which, in general, must be zero, we obtain

$$\frac{\partial g}{\partial y}\eta_1(x) = -\frac{\partial g}{\partial z}\eta_2(x)$$

Thus,

$$\frac{\eta_2(x)}{\eta_1(x)} = -\frac{\partial g/\partial y}{\partial g/\partial z}$$

Also,

$$\frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2}\left[\left(\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'}\right)\eta_1(x) + \left(\frac{\partial f}{\partial z} - \frac{d}{dx}\frac{\partial f}{\partial z'}\right)\eta_2(x)\right] dx$$

Euler's Equation with auxiliary conditions

Combining the last two equations yields,

$$\frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left[\left(\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} \right) - \left(\frac{\partial f}{\partial z} - \frac{d}{dx}\frac{\partial f}{\partial z'} \right) \left(\frac{\partial g/\partial y}{\partial g/\partial z} \right) \right] \eta_1(x)\, dx$$

This latter equation now contains the single arbitrary function $\eta_1(x)$, which is not in any way restricted by the previous Equations. Requiring the above expression to be zero necessitates the expression in the brackets to vanish. Thus we have,

$$\left(\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} \right) \left(\frac{\partial g}{\partial y} \right)^{-1} = \left(\frac{\partial f}{\partial z} - \frac{d}{dx}\frac{\partial f}{\partial z'} \right) \left(\frac{\partial g}{\partial z} \right)^{-1}$$

The left-hand side of this equation involves only derivatives off and g with respect to y and y', and the right-hand side involves only derivatives with respect to z and z'. Because y and z are both functions of x, the two sides of the equation may be set equal to a function of x, which we write as $-\lambda(x)$:

$$\left.\begin{aligned} \frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} + \lambda(x)\frac{\partial g}{\partial y} &= 0 \\ \frac{\partial f}{\partial z} - \frac{d}{dx}\frac{\partial f}{\partial z'} + \lambda(x)\frac{\partial g}{\partial z} &= 0 \end{aligned}\right\}$$

The complete solution to the problem now depends on finding three functions: $y(x)$, $z(x)$, and $\lambda(x)$. But there are three relations that may be used: the two equations above and the constraint equation. Thus, there is a sufficient number of relations to allow a complete solution. Note that here $\lambda(x)$ is considered to be *undetermined* and is obtained as a part of the solution. The function, $\lambda(x)$ is known as a **Lagrange undetermined multiplier**.

Euler's Equation with auxiliary conditions

For the general case of several dependent variables and several auxiliary conditions, we have the following set of equations:

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx}\frac{\partial f}{\partial y_i'} + \sum_j \lambda_j(x)\frac{\partial g_j}{\partial y_i} = 0$$

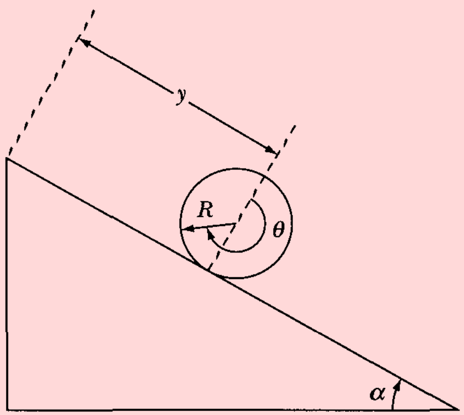
$$g_j\{y_i; x\} = 0$$

If $i = l, 2, \ldots, m$, and $j = l, 2, \ldots, n$, the top equation represents m equations in $m + n$ unknowns, but there are also the n equations of constraint from the second equation. Thus, there are $m + n$ equations in $m + n$ unknowns, and the system is soluble. The constraint equation above is equivalent to the set of n differential equations,

$$\sum_i \frac{\partial g_j}{\partial y_i} dy_i = 0, \qquad \begin{cases} i = 1, 2, \cdots, m \\ j = 1, 2, \cdots, n \end{cases}$$

In problems in mechanics, the constraint equations are frequently differential rather than algebraic equations. Therefore, equations such as the latter above are sometimes more useful than the original constraint equations.

Problem 7. (Rolling Disk)
Consider a disk rolling without slipping on an inclined plane. Determine the equation of constraint in terms of the "generalized coordinates" y and θ.



Euler's Equation with auxiliary conditions

Problem 8. (Rolling Disk)

find the curve $y(x)$ of length l bounded by the x-axis on the bottom that passes through the points $(-a, 0)$ and $(a, 0)$ and encloses the largest area.
The value of the endpoints a is determined by the problem.